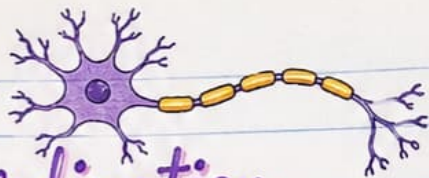


CHAPTER - 18

Neural Control and Coordination

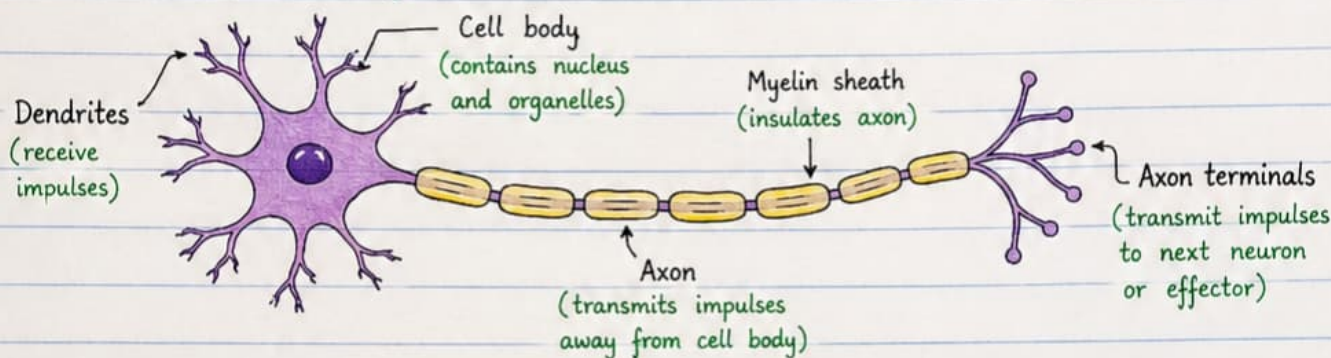


★ The nervous system in animals helps in **quick detection** of changes, **processing of information** and **coordination** of responses.

★ **Neurons** are the basic structural and functional units of nervous system.

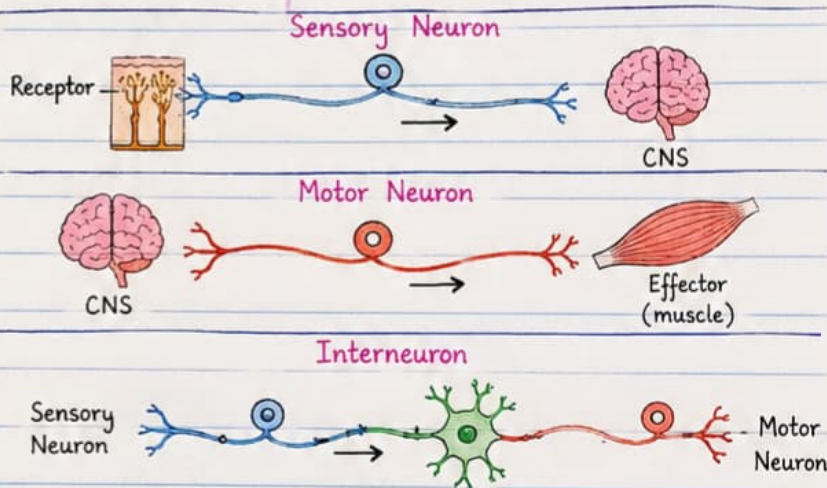
1. NEURON - STRUCTURAL AND FUNCTIONAL UNIT

- Neuron is a specialised cell that conducts **nerve impulses**.
- It consists of **cell body**, **dendrites** and **axon**.



2. TYPES OF NEURONS

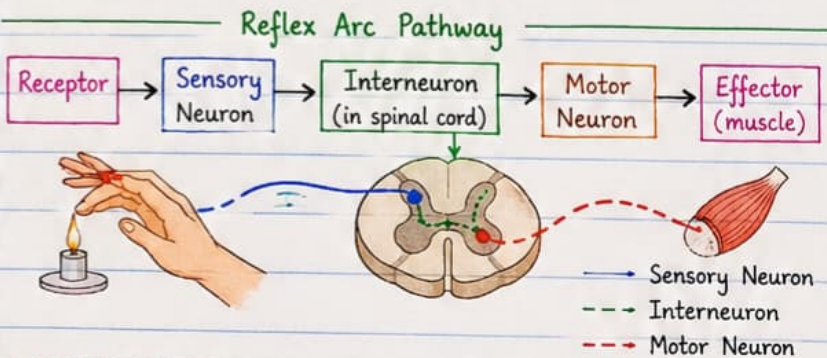
- **Sensory (Afferent) Neurons** - carry impulses from receptors to CNS.
- **Motor (Efferent) Neurons** - carry impulses from CNS to effectors (muscles/glands).
- **Interneurons (Association Neurons)** - present in CNS, connect sensory and motor neurons.



3. REFLEX ACTION

- It is a **rapid, automatic** and **involuntary** response to a stimulus.
- It occurs through **reflex arc**.

Example: Withdrawal of hand when we touch a hot object.

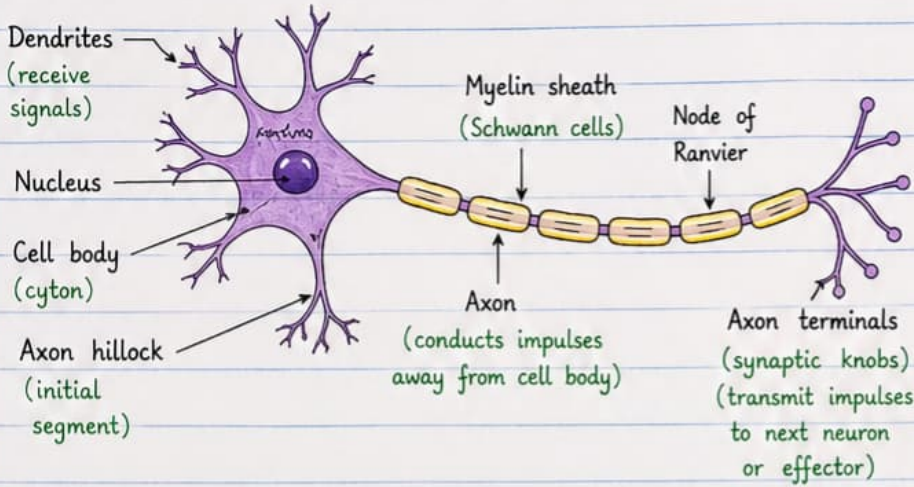


★ Do You Know?

A reflex action takes place even if the brain is not involved directly, which saves time and protects the body.



4. STRUCTURE OF HUMAN NEURON



Do You Know?



Myelin sheath acts as an insulator and increases the speed of nerve impulse conduction.

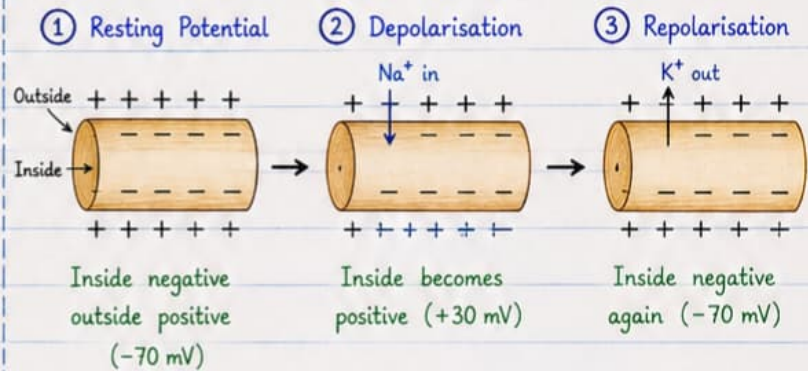


Saltatory conduction occurs in myelinated fibres (jumping from one node to another).

5. NERVE IMPULSE - GENERATION AND CONDUCTION

- At rest, the inside of the neuron is **negatively charged** (-70 mV) and outside is positively charged.
- When a stimulus is applied, ion channels open and Na^+ rushes in $\rightarrow$ **depolarisation** ($+30$ mV).
- Then K^+ moves out $\rightarrow$ **repolarisation** (back to resting potential).
- The impulse moves in **one direction** along the axon.

Events during Nerve Impulse



* Properties of Nerve Impulse

- ① **All or None Law**
Impulse is either generated completely or not at all.



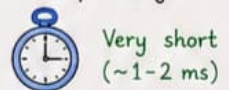
- ② **Non-decremental**
Impulse does not decrease in strength along the axon.



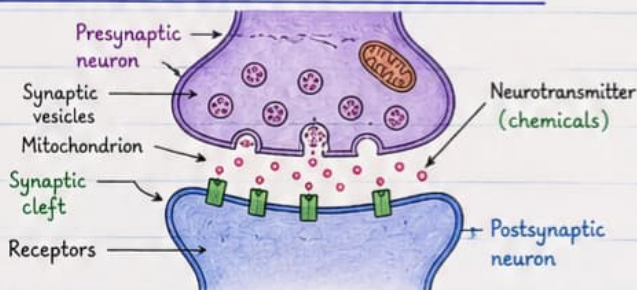
- ③ **Unidirectional**
Impulse travels in one direction from dendrite $\rightarrow$ axon terminals.



- ④ **Refractory Period**
After an impulse, neuron needs a short rest before it can respond again.



6. SYNAPSE - JUNCTION BETWEEN NEURONS



Steps of Synaptic Transmission

- Impulse reaches axon terminal.
- Synaptic vesicles release **neurotransmitters** into synaptic cleft.
- Neurotransmitters diffuse and bind to **receptors** on postsynaptic membrane.
- New impulse is generated in postsynaptic neuron (or response in effector).

Important Neurotransmitters

- Acetylcholine (ACh)** - Muscle contraction, memory
- Dopamine** - Movement, reward, motivation
- Serotonin** - Mood, sleep, hunger
- GABA** - Inhibitory (reduces activity)
- Adrenaline** - Alertness, fight or flight response

Quick Revision

- ✓ Neuron - structural & functional unit of nervous system.
- ✓ Nerve impulse - electrical signal that travels along neuron.
- ✓ Synapse - chemical junction for communication.
- ✓ Nervous system ensures coordination & quick response for survival.

7. HUMAN BRAIN - CONTROL CENTRE

- The brain is the most complex part of CNS.
- It is enclosed in the bony **skull (cranium)** and protected by three **meninges**.
- It is made up of **~86 billion neurons**.
- It receives and interprets information and coordinates all body activities.

Meninges - Protective Membranes

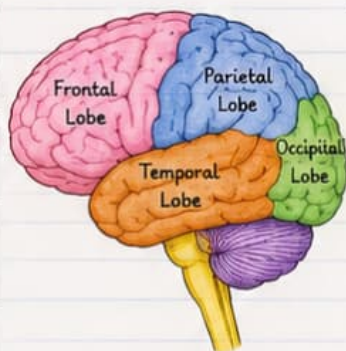
1. **Dura mater** - Tough, outermost membrane.
2. **Arachnoid mater** - Middle, spider-like membrane.
3. **Pia mater** - Thin, delicate inner membrane, closely adherent to the brain.

Cerebrospinal Fluid (CSF)

- CSF is present in ventricles and subarachnoid space.
- It acts as a shock absorber and maintains buoyancy.
- It also supplies nutrients and removes wastes.

LOBES OF CEREBRUM

Lobe	Main Functions
Frontal Lobe	Thinking, decision making, personality, planning, voluntary movement, speech (Broca's area).
Parietal Lobe	Receives and interprets sensory information (touch, pressure, temperature, pain).
Temporal Lobe	Hearing, memory, emotions, understanding language (Wernicke's area).
Occipital Lobe	Visual processing and interpretation.

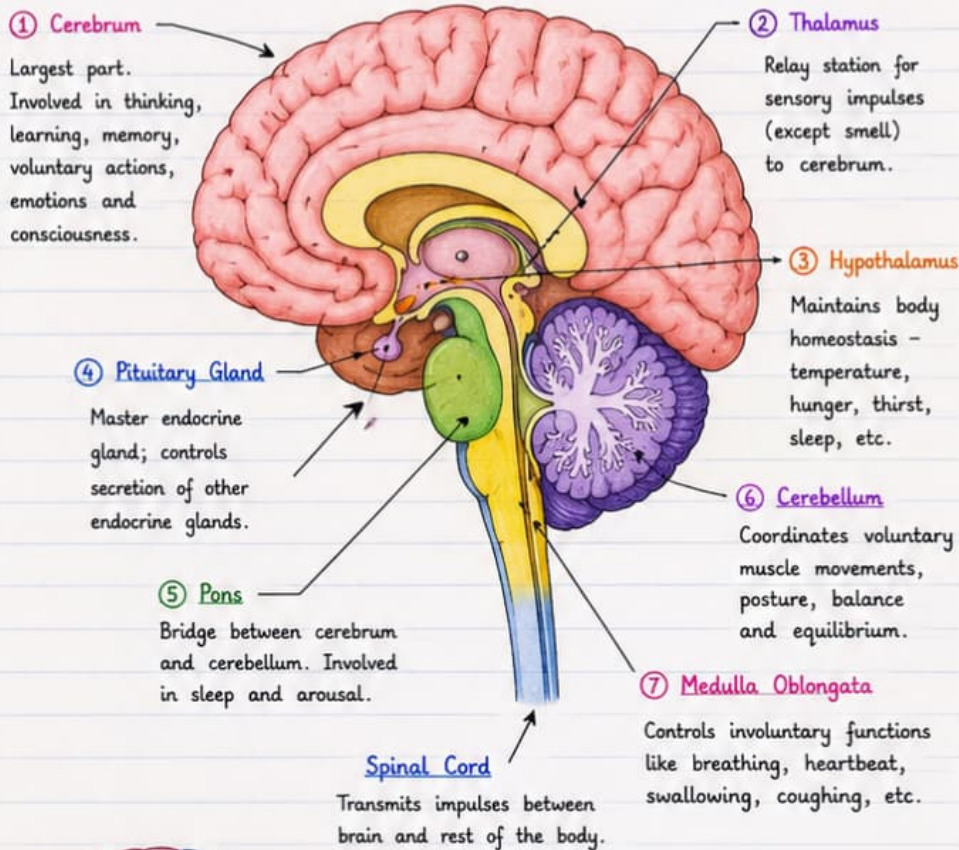


Do You Know?

The human brain weighs about 1.3-1.4 kg in adults, which is only **~2%** of body weight but uses **20%** of the body's total energy!



Human Brain - Major Parts (Sagittal Section)



Functions of Human Brain

- ✓ Receives sensory information.
- ✓ Processes and interprets information.
- ✓ Stores memories and experiences.
- ✓ Initiates and controls voluntary actions.
- ✓ Regulates involuntary activities.
- ✓ Maintains balance, posture and coordination.
- ✓ Supports emotions, intelligence and thinking.

8. CRANIAL NERVES - 12 PAIRS

These nerves arise directly from the brain and help in sensory and motor functions of head, neck and some internal organs.

★ Tip for NEET

Remember: "Some Say Marry Money, But My Brother Says Big Brains Matter More."
(It helps in remembering cranial nerves)

No.	Nerve	Function	No.	Nerve	Function
I	Olfactory	Smell	VII	Facial	Facial expression, taste (ant. 2/3rd tongue)
II	Optic	Vision	VIII	Vestibulocochlear	Hearing, balance
III	Oculomotor	Eye movement	IX	Glossopharyngeal	Swallowing, taste
IV	Trochlear	Eye movement	X	Vagus	Voice, swallowing, internal organs
V	Trigeminal	Facial sensation, chewing	XI	Accessory	Shoulder movement
VI	Abducens	Eye movement	XII	Hypoglossal	Tongue movement